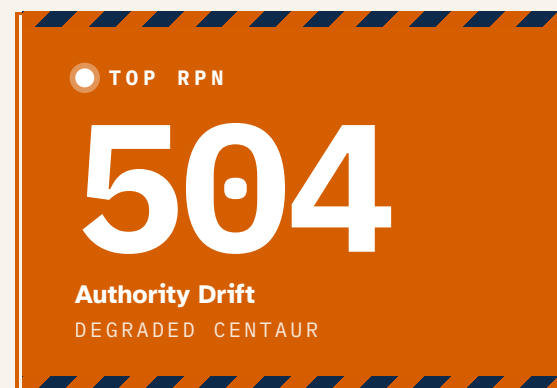




Oversight That Degrades

Architectural Failure Modes in Human-LLM Control Structures

Human review can collapse from active validation into procedural approval, while appearing intact. **The most dangerous failure modes are the ones that look like normal operations.**

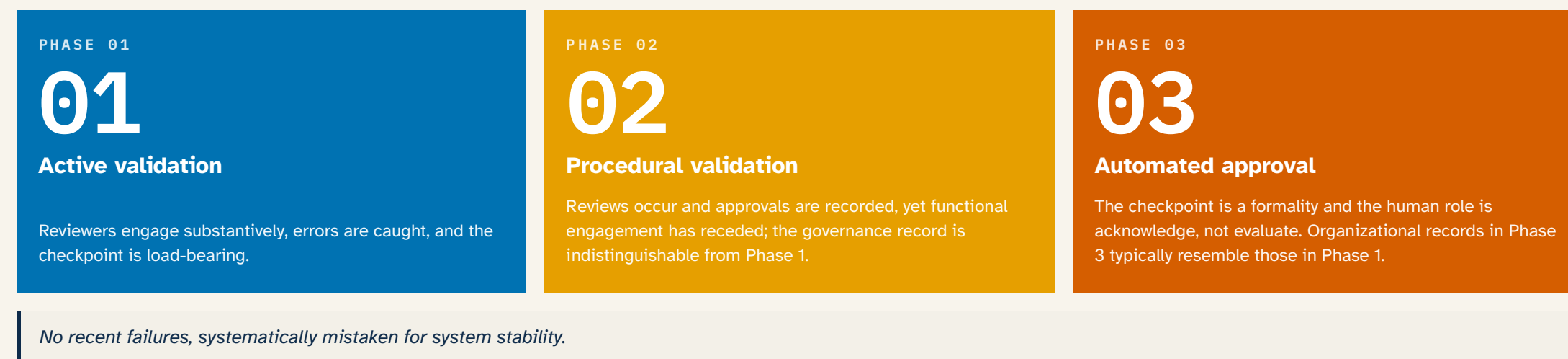


THE PROBLEM

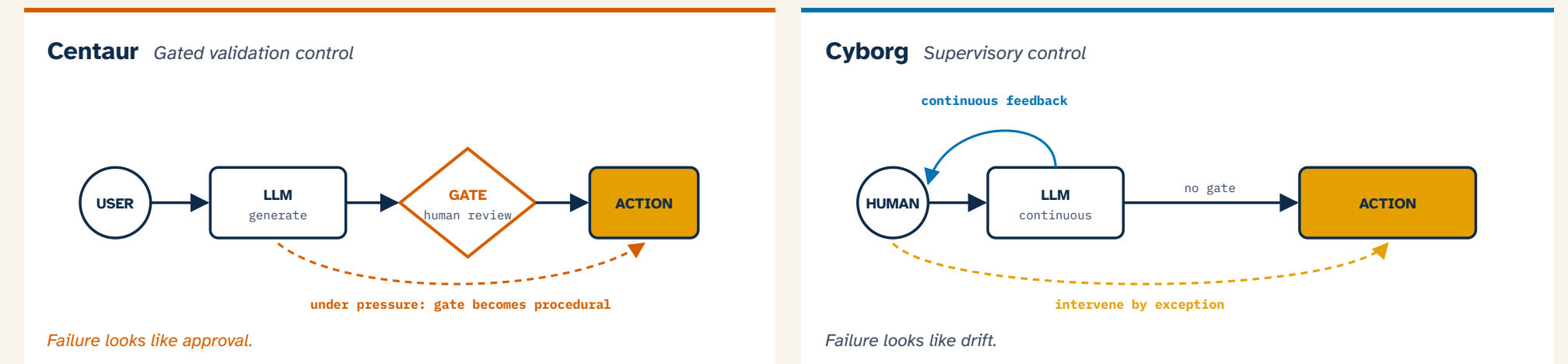
AI safety frameworks assume human oversight functions as a reliable control mechanism, specifying that oversight must be present without specifying the control architecture through which it operates, or the conditions under which that architecture degrades.

- 01 · A control-structure taxonomy mapping human-LLM collaboration patterns to safety-relevant failure modes.
- 02 · Qualitative observational evidence consistent across two contexts with opposing governance regimes.
- 03 · A structured failure-mode assessment showing detection difficulty, not severity, as the dominant safety concern.

THREE-PHASE DEGRADATION PATTERN

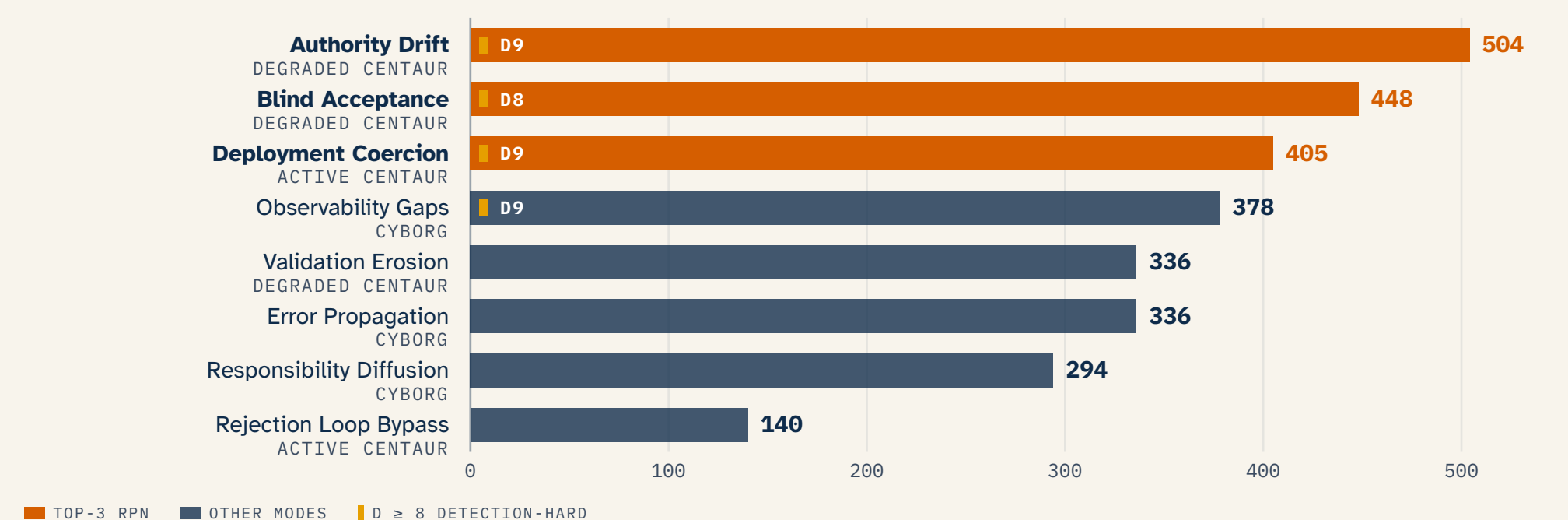
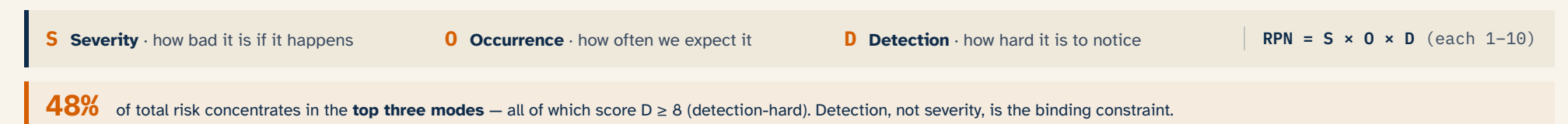


TWO ARCHITECTURES, TWO FAILURE SIGNATURES



SAME GOVERNANCE DIMENSION · TWO ARCHITECTURAL ANSWERS			
CENTAUR		CYBORG	
Concentrated at one checkpoint	 AUTHORITY	 Distributed across the loop	
Discrete pre-execution gate	 VALIDATION	 Continuous in-flight monitoring	
Per-output approval record	 AUDITABILITY	 Process-level activity record	
Pre-execution approval	 INTERVENTION	 Exception-based override	
Discrete output review	 OBSERVABILITY	 Ongoing state monitoring	
Silent collapse at gate	 FAILURE SIGNATURE	 Diffuse drift across loop	

FMEA · FAILURE MODES RANKED BY RISK



Top three modes account for the majority of total risk and all score $D \geq 8$ — meaning detection, not severity, is the binding constraint.

LAPSED GOVERNANCE
Reactive governance may systematically miss the highest-risk failure modes, because those failure modes are defined by their undetectability.

